

Closed-Loop versus Decomposed Go-to-Goal Control of a Unicycle Robot Under Actuation Noise: A Simulation-Based Comparison

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Abstract—Reaching a goal position is the most elementary navigation capability of a mobile robot, yet the choice of the low-level go-to-goal strategy affects both efficiency and robustness. This paper compares two widely used strategies for a kinematic unicycle robot: a proportional closed-loop controller, which continuously and simultaneously regulates the linear and angular velocities as a function of the distance and bearing to the goal, and a decomposed rotate-then-translate baseline, which first aligns the heading and then drives straight, re-aligning whenever the heading error exceeds a threshold. The two controllers are evaluated in a ROS 2-based simulation under three levels of additive Gaussian actuation noise, with a 2×3 full factorial design and $N = 25$ paired trials per condition. The closed-loop controller reaches the goal on average 32% faster at every noise level (paired t -test, $p < 0.001$, $|d| > 3$), while the baseline yields straighter paths (efficiency 0.99 vs. 0.93 in the noise-free case) but degrades roughly twice as fast as noise increases. Contrary to our third hypothesis, the time-to-goal advantage of the closed-loop law does not grow significantly with the noise level ($p = 0.54$). All experiments, data, and figures are fully reproducible through the accompanying Jupyter notebook, which interacts with the ROS 2 experiment node.

Index Terms—mobile robots, unicycle model, go-to-goal control, actuation noise, experimental methodology, ROS 2

I. Introduction and Research Question

Point-to-point navigation — driving a wheeled robot from its current pose to a goal position — underlies virtually every mobile-robot application, from service robotics to warehouse logistics. When a global planner provides a sequence of waypoints, a low-level go-to-goal controller is responsible for actually reaching each of them. Two families of strategies are commonly taught and deployed. The first decomposes the motion: the robot rotates in place until it points towards the goal and then translates along a straight line; the behaviour is simple to implement and to predict, and the resulting path is (nominally) the shortest possible one. The second family uses closed-loop feedback laws that update the linear and angular velocities simultaneously and continuously, typically as proportional functions of the polar coordinates of the goal in the robot frame [1], [2]. Since the two velocity channels are never idle at the same time, the robot keeps advancing while it corrects its heading, at the price of a curved trajectory.

While the closed-loop approach is generally considered superior, the decomposed strategy is still widespread in educational settings and in simple industrial deployments, where its straight-line paths are seen as an advantage — for instance in narrow aisles, where lateral deviation matters more than travel time. Moreover, real robots execute velocity commands imperfectly: gear backlash, wheel slip, discretized motor drivers, and unmodelled dynamics all perturb the executed linear and angular velocities. Intuition suggests that a strategy that corrects errors continuously should tolerate such disturbances better than one that only reacts once a threshold is crossed, but intuition is not evidence. It is therefore legitimate — and, to our knowledge, not yet answered quantitatively in the literature — to ask whether the theoretical advantages of continuous feedback survive, or even grow, when actuation becomes unreliable.

Beyond its immediate object, the study is also designed as an exercise in sound experimental methodology: hypotheses are stated before data collection, all variables are explicitly classified as independent, dependent, or controlled, trials are paired through shared random seeds, and every number and figure in this paper can be regenerated by executing a single Jupyter notebook that drives the ROS 2 experiment node.

Following the compressed PICO structure [9], the research question addressed in this work is:

Does a proportional closed-loop go-to-goal controller improve time-to-goal and path efficiency, compared to a rotate-then-translate baseline, for a unicycle mobile robot navigating to random goals under increasing actuation noise?

Three falsifiable hypotheses were formulated before data collection:

- H1: the proportional controller achieves a lower time-to-goal than the baseline at every noise level;
- H2: the baseline achieves a higher path efficiency in the noise-free case, but its efficiency degrades faster as noise increases;
- H3: the time-to-goal advantage of the proportional

controller increases with the noise level (interaction effect).

II. Literature Analysis

The kinematic control of unicycle-like vehicles is a classical subject. Brockett’s theorem [3] establishes that no smooth, time-invariant, pure-state feedback can asymptotically stabilize a nonholonomic vehicle to a full pose, which historically motivated two research directions: discontinuous or time-varying laws for pose stabilization [4], and smooth laws for the easier position-only stabilization problem considered here. Within the second direction, Aicardi et al. [1] introduced the polar coordinate transformation and the Lyapunov-based proportional law on which our “proposed” controller is modelled: expressing the goal through its distance ρ and bearing error α , the simple feedback $v = k_\rho \rho$, $\omega = k_\alpha \alpha$ yields global convergence with smooth, arc-shaped trajectories. This law, popularized by the textbook of Siegwart et al. [2], remains the standard entry point for go-to-goal control. In the closely related trajectory-tracking setting, Kanayama et al. [5] proposed a stable tracking scheme whose linearization is again a proportional feedback on the posture error, reinforcing the centrality of proportional laws in this domain. Later refinements focused on graceful motion: Park and Kuipers [6] derived a smooth control law with curvature-bounded trajectories for differential-drive robots, confirming that continuous feedback produces faster and smoother motion than decomposed schemes, though without a systematic statistical comparison under actuation noise. A common trait of these control-theoretic works is that performance is demonstrated by convergence proofs and a handful of illustrative trajectories, not by repeated randomized experiments: robustness to stochastic actuation error is argued qualitatively, when discussed at all.

A second body of literature addresses obstacle-aware local navigation. The dynamic window approach [7] and its descendants, up to the ROS 2 Nav2 stack [8], embed goal-seeking behaviour into velocity-space optimization. These controllers presuppose exactly the kind of elementary go-to-goal capability studied here, but their complexity confounds the analysis of the low-level strategy itself; benchmarking papers about Nav2 typically compare whole planning pipelines rather than the underlying motion primitives. Interestingly, Nav2 itself ships a “rotate-to-heading, then drive” behaviour in its rotation shim controller, precisely because pure closed-loop controllers may start driving in the wrong direction when the initial heading error is large — evidence that the decomposed strategy is not merely a pedagogical strawman but a design option that practitioners actively choose.

Finally, a growing literature on experimental methodology in robotics [9], [10] stresses replicability: experiments should isolate independent variables, report aggregate statistics over repeated randomized trials, and publish code and data. Interactive notebooks are increasingly

adopted as the vehicle for such reproducible workflows [11].

Gap. Across these three threads, the closed-loop and decomposed go-to-goal strategies are either assumed (navigation stacks), analysed only in the noise-free deterministic setting (control-theoretic works), or compared qualitatively. To the best of our knowledge, no work reports a controlled, statistically analysed comparison of the two strategies under increasing actuation noise, with paired randomized trials and public code. This paper fills that small but well-defined gap, and does so with a fully reproducible ROS 2 + notebook pipeline.

III. Methodological Proposal

A. Robot and noise model

The robot is modelled as a kinematic unicycle with state (x, y, θ) and inputs (v, ω) , integrated with explicit Euler at $\Delta t = 0.05$ s:

$$x^+ = x + v \cos \theta \Delta t, \quad y^+ = y + v \sin \theta \Delta t, \quad \theta^+ = \theta + \omega \Delta t. \quad (1)$$

Actuation imperfection is modelled as additive zero-mean Gaussian noise on the executed velocities at every step:

$$v = v_{\text{cmd}} + \mathcal{N}(0, \sigma V_{\text{max}}), \quad \omega = \omega_{\text{cmd}} + \mathcal{N}(0, \sigma W_{\text{max}}), \quad (2)$$

where $\sigma \in \{0.0, 0.1, 0.3\}$ is the dimensionless noise level and $V_{\text{max}} = 2$ m/s, $W_{\text{max}} = 2$ rad/s are the saturation limits. Executed linear velocities are clipped at zero (no reverse motion).

B. Controllers

Both controllers observe the goal through its polar coordinates in the robot frame: the distance $\rho = \|(x_g - x, y_g - y)\|$ and the bearing error $\alpha = \text{atan2}(y_g - y, x_g - x) - \theta$, wrapped to $(-\pi, \pi]$. The proportional closed-loop controller (proposed) is the classical polar law [1] with a cosine gating that slows the robot down when badly oriented:

$$v = k_\rho \rho \max(\cos \alpha, 0), \quad \omega = k_\alpha \alpha, \quad (3)$$

with $k_\rho = 1.2$, $k_\alpha = 3.0$. The gating term preserves the convergence properties of the original law while preventing the robot from driving forward when the goal is behind it. The rotate-then-translate baseline implements the decomposed strategy as a two-state machine: it rotates in place at $\omega = \pm 1.5$ rad/s while $|\alpha| > 0.2$ rad, and otherwise translates at a constant $v = 1$ m/s with $\omega = 0$; whenever noise pushes the heading error back above the threshold, it stops and re-aligns. Both outputs are saturated to $[0, V_{\text{max}}]$ and $[-W_{\text{max}}, W_{\text{max}}]$. The gains were fixed once, by hand, so that both controllers use comparable fractions of the available velocity envelope; neither was tuned per condition, which keeps the comparison fair but is acknowledged as a limitation in Sec. VI.

C. Software architecture

The experiment is implemented as a ROS 2 (Jazzy) Python package, `goal_nav_experiment`¹. A dedicated experiment node receives JSON trial requests on the topic `/experiment/run` (controller name, noise level, seed), simulates the trial, publishes the resulting metrics on `/experiment/result`, and appends every trial to a CSV log. A Jupyter notebook acts as the ROS 2 client: it launches the node, requests single interactive trials through ipywidgets controls, runs the full experimental campaign, and produces all statistics and figures reported here. Because every trial is driven by a seeded random generator, the entire study is exactly reproducible [11].

The simulated experiment makes the following explicit assumptions: (i) purely kinematic motion (no inertia, slip, or actuator dynamics beyond the noise term); (ii) perfect localization (the controllers read the true pose); (iii) an obstacle-free 11×11 m arena; (iv) noise that is additive, Gaussian, zero-mean, and white. These assumptions isolate the independent variable of interest — the control strategy — from confounding factors.

IV. Experiment Design

A. Variables

Independent variables: controller type (proportional, rotate-then-translate) and actuation noise level ($\sigma \in \{0.0, 0.1, 0.3\}$), in a 2×3 full factorial design. Dependent variables: time-to-goal [s]; path length [m]; path efficiency, defined as the ratio between the distance actually covered towards the goal and the travelled path length ($1 = \text{straight line}$); success rate (goal reached within the time-out). Control variables: arena size, start pose (5.5, 5.5, 0), velocity limits, integration step, goal tolerance (0.15 m), timeout (60 s), controller gains, and the goal distribution (uniform in $[1, 10]^2$, at least 3 m from the start).

B. Protocol

For every condition, $N = 25$ trials are executed. The same 25 seeds are reused across all six conditions, so that both controllers face exactly the same sequence of goals under the same noise realization schedule: the design is fully paired, which increases statistical power and justifies the use of the paired t -test. The number of trials was chosen after a small pilot (5 runs per condition) showed low variability and a very large effect size for time-to-goal. In total, $2 \times 3 \times 25 = 150$ trials are executed automatically by the notebook in a few seconds of wall-clock time.

Confounding factors are handled as follows. Environmental variability is eliminated by construction (the arena is fixed and empty); goal-position variability, which strongly affects all metrics, is neutralized by the paired seeding; the stochastic noise realization is the only residual source of within-pair variability, and it is averaged out by

¹Code, executed notebook, and data: https://github.com/Tawakoll/RT2_Final_Project

TABLE I
Aggregate results (mean $\pm$ std over $N = 25$ trials).

σ	Controller	Time-to-goal [s]	Path efficiency
0.0	Proportional	3.67 ± 0.50	0.927 ± 0.046
0.0	Rotate-translate	5.40 ± 0.89	0.987 ± 0.002
0.1	Proportional	3.69 ± 0.57	0.922 ± 0.052
0.1	Rotate-translate	5.39 ± 0.92	0.975 ± 0.018
0.3	Proportional	3.52 ± 0.57	0.910 ± 0.066
0.3	Rotate-translate	5.34 ± 1.02	0.956 ± 0.049

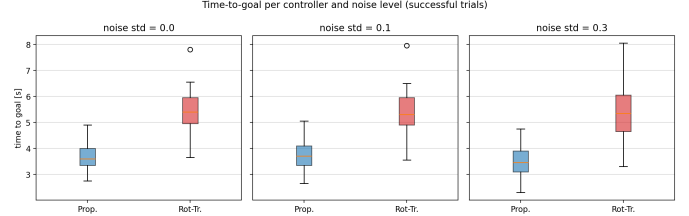


Fig. 1. Distribution of the time-to-goal for the two controllers at the three noise levels (box plots over $N = 25$ trials).

the repeated trials. Because the simulation is deterministic given the seed, re-running the campaign reproduces the dataset bit for bit — failures, if any, are included in the analysis rather than discarded.

C. Statistical analysis

For each metric and noise level, the two controllers are compared with a two-tailed paired t -test at significance level $\alpha = 0.05$; the Wilcoxon signed-rank test is used as a non-parametric robustness check, and Cohen’s d on the paired differences quantifies the effect size. H3 is tested by comparing the paired time-to-goal differences at $\sigma = 0.3$ against those at $\sigma = 0.0$ with a Welch two-sample t -test.

V. Results and Visualization

A. Aggregate metrics

All 150 trials succeeded, so the success rate saturates at 100% in every condition and is not analysed further. Table I reports the mean and standard deviation of the two main metrics.

Fig. 1 shows the time-to-goal distributions: the proportional controller is not only faster on average (≈ 1.7 – 1.8 s, i.e. about 32%) but also less dispersed. Fig. 2 summarizes both metrics as a function of noise: time-to-goal is remarkably flat for both controllers, while path efficiency exhibits the crossing pattern predicted by H2 — the baseline starts near the ideal value 0.99 and loses 0.031 as σ grows to 0.3, whereas the proportional controller starts lower (0.93) and loses only 0.017.

B. Hypothesis tests

Table II reports the paired comparisons. For time-to-goal the proportional controller is significantly faster at every noise level ($p < 10^{-13}$, $|d| \in [3.3, 4.1]$), confirming H1. For path efficiency the baseline is significantly better

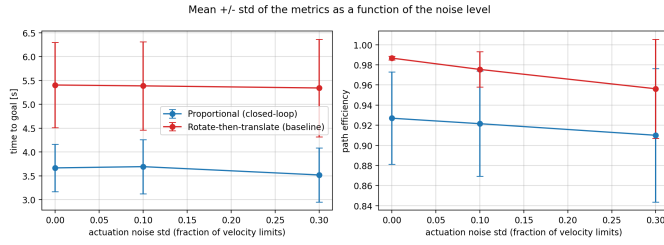


Fig. 2. Mean $\pm$ std of time-to-goal (left) and path efficiency (right) as a function of the actuation noise level.

TABLE II

Paired comparisons (proportional – baseline), $N = 25$ pairs.

Metric / σ	Mean diff.	t	p (paired)	Cohen's d
Time 0.0	-1.74 s	-20.4	< 0.001	-4.07
Time 0.1	-1.69 s	-20.1	< 0.001	-4.03
Time 0.3	-1.82 s	-16.5	< 0.001	-3.30
Effic. 0.0	-0.060	-6.6	< 0.001	-1.32
Effic. 0.1	-0.054	-6.8	< 0.001	-1.37
Effic. 0.3	-0.046	-7.5	< 0.001	-1.51

at every noise level ($p < 10^{-6}$), confirming the first part of H2; its degradation with noise (-0.031) is roughly twice that of the proportional law (-0.017), consistent with the second part. The Wilcoxon tests agree with all paired t -tests ($p < 0.001$), so the conclusions do not depend on the normality assumption.

Regarding H3, the mean paired advantage of the proportional controller changes only from -1.74 s at $\sigma = 0.0$ to -1.82 s at $\sigma = 0.3$; the Welch test on the paired differences yields $t = -0.62$, $p = 0.54$. H3 is therefore not supported: within the tested range, the advantage of continuous feedback is large but essentially constant with respect to actuation noise.

C. Qualitative evaluation

Fig. 3 overlays the trajectories obtained for one representative goal. The proportional controller produces a single smooth arc whose shape is barely affected by noise. The baseline path is straighter — which explains its higher efficiency — but under noise it exhibits small kinks where the heading error crossed the 0.2 rad threshold and the robot stopped to re-align; each of those stops costs time without adding much path length, which explains why the baseline loses efficiency faster than it loses time.

VI. Discussion and Limitations

The experiment confirms the textbook intuition with quantitative evidence: continuous proportional feedback dominates the decomposed strategy for time-optimal goal reaching (very large effect, $|d| > 3$), while the decomposed strategy retains a genuine, statistically significant advantage in path efficiency. The practical implication is that the choice is a true trade-off only when path straightness matters more than time (e.g., narrow corridors); in all other cases the closed-loop law should be preferred. The

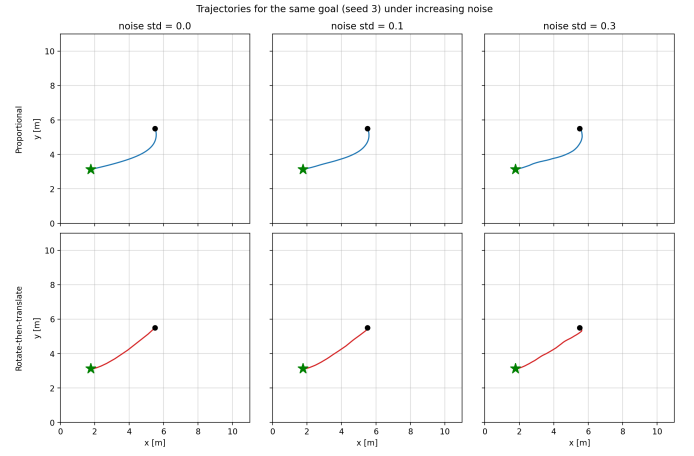


Fig. 3. Qualitative comparison: trajectories towards the same goal (seed 3) for the proportional controller (top, blue) and the rotate-then-translate baseline (bottom, red) under increasing noise. Star: goal; dot: start.

magnitude of the gap is worth emphasizing: a 32% reduction in time-to-goal, obtained by changing a few lines of controller code, is larger than the improvements typically reported for far more sophisticated interventions on the planning side of the navigation stack. Conversely, the efficiency cost of the closed-loop law is bounded ($\leq 6\%$ longer paths on average) and shrinks as noise increases, so the baseline's only advantage erodes precisely in the conditions where robustness matters.

The variability of the two controllers also differs in an interpretable way. The time-to-goal standard deviation of the baseline is roughly twice that of the proportional law at every noise level (Table I), because the number of stop-and-rotate phases — each costing a discrete amount of time — varies from trial to trial. In applications where predictable arrival times matter (e.g., fleet scheduling), this dispersion penalty is an additional argument for continuous feedback.

The rejection of H3 is instructive. We expected stop-and-rotate phases to multiply under noise, widening the time gap. Instead, because the injected noise is zero-mean and white, its net effect over a trial largely averages out, and the baseline's threshold mechanism absorbs moderate heading disturbances without triggering extra rotations. A noise model with bias (e.g., asymmetric wheel slip) or a narrower alignment threshold would likely produce the hypothesized interaction; this is a concrete direction for future work. As the methodology literature emphasizes, an experiment whose outcome was certain in advance would have been pointless — negative results are results.

Threats to validity. Internal validity is protected by the paired design, the seeded randomization, and the strict control of all remaining variables. External validity is limited by the simulation assumptions: the kinematic model ignores dynamics and slip; localization is perfect, so the comparison captures actuation noise only; the arena is

obstacle-free; the gains of both controllers are fixed rather than optimized per condition, and the tuning inevitably influences the absolute numbers; and the success metric saturated (100%), so robustness differences beyond the tested noise range remain unobserved. Finally, the noise range itself ($\sigma \leq 0.3$) was chosen to represent moderate actuation imperfection; harsher or biased noise may change the picture.

VII. Conclusion

This paper presented a controlled, fully reproducible comparison between a proportional closed-loop go-to-goal controller and a rotate-then-translate baseline for a unicycle robot under actuation noise, implemented as a ROS 2 node driven by a Jupyter notebook. The closed-loop law is about 32% faster at every noise level with a very large effect size (H1 confirmed); the baseline yields straighter paths but loses efficiency about twice as fast as noise grows (H2 confirmed); and the time advantage of the closed-loop law does not increase with noise in the tested range (H3 rejected). Future work includes biased and colored noise models, sensor (localization) noise, obstacle-cluttered environments, and per-condition gain tuning.

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